

WEYL-CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL-RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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1. INTRODUCTION

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$, $N \geq 2$, and let G be the corresponding finite reflection group. Let $\kappa : R \rightarrow \mathbb{R}_+$ be G -invariant multiplicity function. Let $R_{\text{Dunkl},j}$ be the j -th Dunkl-Riesz transform corresponding to R and κ . Let w be the Dunkl weight corresponding to R and κ . Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$.*

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- (i) If $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (ii) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_\kappa^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (iii) For every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ and for every $1 \leq j \leq N$, there exists a limit

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \text{write expression}.$$

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $|\cdot|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called normalised if $|v|^2 = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R .

Let $\kappa : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

Define $d(x, y) = \min_{g \in G} |x - gy|$. Recall from [8, Chapter VII, proof of Theorem 2.12] that $d(x, y) = |x - y|$ iff x and y belong to the same chamber.

Define the Dunkl weight on $\mathbb{R}^N$ by

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)}, \quad x \in \mathbb{R}^N.$$

Dunkl measure $w(x)dx$ is also denoted by w .

Let $B(x, r) := \{y \in \mathbb{R}^N : |x - y| < r\}$. Let $\mathbf{N} := N + \sum_{v \in R} \kappa(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, \ t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^N \prod_{v \in R} (|\langle x, v \rangle| + r)^{\kappa(v)},$$

where the implicit constant only depends on N, R and κ . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{\kappa(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} = T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \dots, N$.

Operators $U_g : f \rightarrow f \circ g^{-1}$, $g \in G$, are unitary both in $L_2(\mathbb{R}^N)$ and in $L_2(\mathbb{R}^N, w(x)dx)$. Obviously, $g \rightarrow U_g$ is a unitary representation of the group G . To lighten the notations, we write U_v instead of U_{σ_v} for $v \in R$.

2.2. Weyl chamber as simplicial cone.

Definition 2.1. A connected component of the set

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

Theorem 2.2. Let Γ be a chamber. There exists a linear isomorphism $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$ (with $N_1 + N_2 = N$).

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [9, 11, 3, 4, 10, 12, 14, 13]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [9, Section 13] for a complete proof).

Notation 2.3. Let

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube in $\mathbb{D}_k$ is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube $Q \in \mathbb{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Definition 2.4 and Lemma 2.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

Definition 2.4. Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_Q\}_{Q \in \mathbb{D}}$ is called (r, c) -NWO if

- (i) $\text{supp}(e_Q) \subseteq cQ$ for every $Q \in \mathbb{D}$;
- (ii) $\|e_Q\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} \leq \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}}$ for every $Q \in \mathbb{D}$;

Lemma 2.5. *Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $0 < p < \infty$. *Should we also ask that our metric-measure space is of homogeneous type?* Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} &\leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, \mathbf{w}(x)dx)) \right) &\leq \\ &\leq C_{\mathbf{w},r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left(\{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. $\square$

Lemma 2.6. *Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $1 < p < \infty$. *Should we also ask that our metric-measure space is of homogeneous type?* Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\left\| \left\{ \langle A e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{p,q}(\mathbb{D})} \leq c_{\mathbf{w},r_1,c_1,r_2,c_2,p,q} \|A\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}.$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. $\square$

2.4. Principal symbol and crossed products. Let Π be the C^* -subalgebra in $B(L_2(\mathbb{R}^N))$ generated by M_f , $f \in (C_0 + \mathbb{C})(\mathbb{R}^N)$ and $h(\frac{\nabla}{\sqrt{\Delta}})$, $h \in C(\mathbb{S}^{N-1})$. Note that every compact operator is contained in Π . There exists a surjective $*$ -homomorphism

$$\text{sym} : \Pi \rightarrow (C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$$

such that $\ker(\text{sym})$ consists exactly of all compact operators.

We have a natural action α of G on $B(L_2(\mathbb{R}^N))$ and, therefore, on Π — the C^* -algebra of operators admitting principal symbol. Namely,

$$\alpha(g) : T \rightarrow U_g T U_g^{-1}, \quad T \in B(L_2(\mathbb{R}^N)), \quad g \in G.$$

Define the action β of G on $(C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$ by setting

$$(\beta(g))(f \otimes h) = (f \circ g^{-1}) \otimes (h \circ g^{-1}).$$

The same action extends to von Neumann algebra $L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})$. We have an equivariance relation

$$\text{sym}(\alpha(g)(T)) = (\beta(g))(\text{sym}(T)).$$

Clearly, sym is surjective, $\ker(\text{sym})$ is α -invariant. Hence, there exists a natural $*$ -homomorphism

$$\pi \rtimes G : \Pi \rtimes_\alpha G \rightarrow ((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

We have an inclusion

$$((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_{\beta} G \subset (L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G.$$

Hence,

$$\pi \rtimes G : \Pi \rtimes_{\alpha} G \rightarrow (L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G.$$

Next, the standard trace $\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}}$ on $L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})$ is β -invariant. Hence, there exists a natural trace

$$\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \right) \rtimes G$$

on $(L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G$.

2.5. Birman-Solomyak lemmas.

Lemma 2.7. *Let $A \in \mathcal{L}_{p,\infty}$ and $B \in (\mathcal{L}_{p,\infty})_0$. If*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A) = c,$$

then also

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A + B) = c.$$

Lemma 2.8. *Let $\{A_k\}_{k \geq 1} \subset \mathcal{L}_{p,\infty}$. If*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A_k) = c_k, \quad k \geq 1,$$

and if $A_k \rightarrow A$ in $\mathcal{L}_{p,\infty}$ as $k \rightarrow \infty$, then the following limits exist and are equal

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A) = \lim_{k \rightarrow \infty} c_k.$$

3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

Theorem 3.1. *Let Γ be a chamber. For all $x \in \Gamma$ and for all $r > 0$, we have*

$$w(B(x, r) \cap \Gamma) \geq c_{\kappa} w(B(x, r)).$$

In particular, $(\Gamma, |\cdot|, w(x)dx)$ is a doubling metric measure space.

Notation 3.2. *Let $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$. Set*

$$w_{\theta}(x) = \prod_{A \subset \{1, \dots, N_1\}} \left(\sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

Lemma 3.3. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq c_{N, \theta}^{(1)} r^{N_1} w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \subset \left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} & \int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq \\ & \leq \int_{\left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_{\theta}(y_1, \dots, y_{N_1}) dy = \end{aligned}$$

$$\begin{aligned}
&= \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_\theta(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\
&\leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\
&= w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\
&\leq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}.
\end{aligned}$$

□

Lemma 3.4. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq c_{N, \theta}^{(2)} r^N w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$x + [\frac{r}{2N}, \frac{r}{N}]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned}
&\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\
&\geq \int_{x + [\frac{r}{2N}, \frac{r}{N}]^N} w_\theta(y_1, \dots, y_{N_1}) dy = \\
&= \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta(u) du \times (\frac{r}{2N})^{N_2} \geq \\
&\geq \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times (\frac{r}{2N})^{N_2} = \\
&= w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \geq \\
&\geq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}.
\end{aligned}$$

□

Lemma 3.5. *There exists $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let A_v be the set of those $1 \leq k \leq N_1$ with $((\Phi^{-1})^*(v))_l$ being strictly positive. We, obviously, have

$$c_{\Phi, v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi, v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi, v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi, v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. $\square$

Proof of Theorem 3.1. The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. $\square$

4. ROCHBERG-SEMMES BASED ESTIMATE

Theorem 4.1. *Let $N \geq 3$, $1 \leq N_1 \leq N$ and $N_2 = N - N_1$. Let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Let $\mathbf{w} \in A_2(\Omega)$ be such that*

$$\sup_{x \in Q} \mathbf{w}(x) \leq c_{\mathbf{w}} \frac{\mathbf{w}(Q)}{m(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Should we also ask that our metric-measure space is of homogeneous type? Suppose that $K : \Omega \times \Omega \rightarrow \mathbb{C}$ satisfies

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha| - |\beta|_1}}{\mathbf{w}(B(x, |x - y|))}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator on $L_2(\Omega, \mathbf{w}(x)dx)$ with integral kernel K . For every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}} \|f\|_{L_N(\Omega)}.$$

Lemma 4.2. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every x , there exists a unique $Q \in \mathbb{D}_j$ containing x . For this Q , we have $\ell(Q) = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x - y)) = \sum_{j \in \mathbb{Z}} \left(\sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x - y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right). \end{aligned}$$

If $x \in Q$ and $\psi(\frac{x - y}{\ell(Q)}) \neq 0$, then

$$x \in Q, \quad |x - y|_\infty \leq \ell(Q).$$

We obviously have $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|x - y|_\infty \leq \ell(Q)$. Thus, $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$. In other words, $y \in 3Q$. Thus,

$$\chi_Q(x)\psi\left(\frac{x-y}{\ell(Q)}\right) = \chi_Q(x)\chi_{3Q}(y)\psi\left(\frac{x-y}{\ell(Q)}\right).$$

This completes the proof. $\square$

Lemma 4.3. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y),$$

where

(i) $e_{k, l, Q}$ is supported in Q and $f_{k, l, Q}$ is supported in $3Q$ for every $Q \in \mathbb{D}$ with $Q \subset \Omega$;

(ii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|e_{k, l, Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)};$$

(iii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|f_{k, l, Q}\|_{L_\infty(\mathbb{R}^N)} \leq \frac{c_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)};$$

Proof. For every $x, y \in \Omega$, we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \end{aligned}$$

For every $Q \in \mathbb{D}$ with $3Q \subset \Omega$, define the map

$$K_Q : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + a\ell(Q), c_Q + b\ell(Q))\psi(a - b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

If $Q \in \mathbb{D}$ is such that $3Q \not\subset \Omega$, then the definition of K_Q is the same, a also runs through $[-\frac{1}{2}, \frac{1}{2}]^N$, but the parallelepiped b runs through is slightly smaller. **to co-authors: if you wish, specify the domain of b also.** In what follows, we lighten the notations by pretending that b is always defined on $[-\frac{3}{2}, \frac{3}{2}]^N$.

It follows from the assumption on K that

$$\|K_Q\|_{C^{2N+2}([- \frac{3}{2}, \frac{3}{2}]^N \times [- \frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

Hence, there exists an extension $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2, 2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k, l \in \mathbb{Z}^N} \text{coef}_{k, l, Q} e^{i\langle k, a \rangle} e^{i\langle l, b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k, l, Q}|^2 \leq \left(\frac{c_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k, l, Q}| \leq \frac{c_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k, l, Q}(x) = f(x) \chi_Q(x) e^{i\langle k, \frac{x - c_Q}{\ell(Q)} \rangle},$$

$$f_{k, l, Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k, l, Q} \cdot \chi_{3Q}(y) e^{i\langle l, \frac{y - c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x) K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y).$$

Obviously,

$$\text{supp}(e_{k, l, Q}) \subset Q, \quad \text{supp}(f_{k, l, Q}) \subset 3Q,$$

$$\|e_{k, l, Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)},$$

$$\|f_{k, l, Q}\|_{L_\infty(\mathbb{R}^N)} = (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k, l, Q}| \leq \frac{c_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

□

Lemma 4.4. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}, r} \left\| \left\{ m(Q)^{\frac{1}{N} - \frac{1}{r}} \|f\|_{L_r(Q)} \right\}_{Q \in \mathbb{D}} \right\|_{N, \infty}.$$

Proof. Let $A_{k, l}$ be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.3, we have

$$M_f V = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k, l}.$$

By the triangle inequality,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, w_\theta(x)dx))} \leq \\ & \leq \frac{N}{N-1} \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k, l}\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, w_\theta(x)dx))}. \end{aligned}$$

By Lemma 4.3, the sequences

$$\left\{ \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}} \frac{e_{k, l, Q}}{\|f\|_{L_r(Q, \mathbf{w}(x)dx)}} \right\}_{Q \in \mathbb{D}}, \quad \left\{ \frac{c_{K, \theta}^{-1} \mathbf{w}(Q)^{\frac{1}{2}}}{\ell(Q)} f_{k, l, Q} \right\}_{Q \in \mathbb{D}}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 2.4. Using Lemma 2.5, we conclude that

$$\begin{aligned} & \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ & \leq c_{\theta,r,1,\infty,3,N} c_{K,\theta} \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}_+^N, \mathbf{w}(x)dx))} \leq \\ & \leq \frac{N}{N-1} c_{\theta,r,1,\infty,3,N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \\ & \times \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{N,\infty}. \end{aligned}$$

Obviously,

$$\mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \leq \left(\frac{1}{\mathbf{w}(Q)} \cdot \sup_{x \in Q} \mathbf{w}(x) \right)^{\frac{1}{r}} \times \|f\|_{L_r(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof. $\square$

Lemma 4.5. *Let $1 < s < \infty$. We have*

$$\left\| \left\{ m(Q)^{\frac{1}{s}-1} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}} \right\|_{s,\infty} \leq 2s' \|F\|_{L_s(\mathbb{R}^N)}, \quad F \in L_s(\mathbb{R}^N).$$

Proof. Denote $s' = \frac{s}{s-1}$. For a finite set $E \subset \mathbb{D}$, denote

$$G_A = \sum_{Q \in E} m(Q)^{-\frac{1}{s'}} \chi_Q.$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in E} Q$ we may choose a cube $Q_x \in E$ of minimal volume such that $x \in Q_x$. Every $Q \in E$ containing x is, therefore, a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation, it follows that

$$\begin{aligned} G_E(x) &= \sum_{\substack{Q \in E \\ x \in Q}} m(Q)^{-\frac{1}{s'}} \leq m(Q_x)^{-\frac{1}{s'}} \sum_{k=0}^{\infty} 2^{-\frac{kN}{s'}} = \frac{1}{1 - 2^{-\frac{N}{s'}}} m(Q_x)^{-\frac{1}{s'}} \leq \\ & \leq \frac{1}{1 - 2^{-\frac{1}{s'}}} m(Q_x)^{-\frac{1}{s'}} \leq 2s' m(Q_x)^{-\frac{1}{s'}} \leq 2s' \sup_{Q \in E} m(Q)^{-\frac{1}{s'}} \chi_Q(x). \end{aligned}$$

Consequently,

$$\begin{aligned} \|G_E\|_{L_{s'}(\mathbb{R}^N)}^{s'} &\leq (2s')^{s'} \int_{\mathbb{R}^N} \sup_{Q \in E} m(Q)^{-1} \chi_Q(x) dx \leq \\ &\leq (2s')^{s'} \int_{\mathbb{R}^N} \sum_{Q \in E} m(Q)^{-1} \chi_Q(x) dx = (2s')^{s'} |E|. \end{aligned}$$

Hence,

$$\|G_E\|_{L_{s'}(\mathbb{R}^N)} \leq 2s' |E|^{\frac{1}{s'}}.$$

By Hölder's inequality,

$$\sum_{Q \in E} \left(m(Q)^{-\frac{1}{s'}} \int_Q |F(x)| dx \right) = \int_{\mathbb{R}^N} |F(x)| G_E(x) dx \leq 2s' \|F\|_{L_s(\mathbb{R}^N)} \cdot |E|^{\frac{1}{s'}}.$$

In other words,

$$\sup_{\substack{E \subset \mathbb{D} \\ |E| < \infty}} |E|^{-\frac{1}{s'}} \left(\sum_{Q \in E} \left(m(Q)^{-\frac{1}{s'}} \int_Q |F(x)| dx \right) \right) \leq 2s' \|F\|_{L_s(\mathbb{R}^N)}.$$

Appealing to the definition of $l_{s,\infty}$, we complete the proof. $\square$

Proof of Theorem 4.1. Take $r \in (2, N)$. The assertion follows by combining Lemmas 4.4 and 4.5. $\square$

5. PROOF OF SPECIAL CBIKEL ESTIMATE IN DUNKL SETTING

Lemma 5.1. *For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^N$, for every $t > 0$, we have*

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa, \alpha, \beta} t^{-\frac{|\alpha|_1 + |\beta|_1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp\left(-\frac{c_\kappa}{t} d(x, y)^2\right).$$

Proof. This is (slightly weakened form of) Theorem 3.1 in [2]. $\square$

Lemma 5.2. *Suppose $s + N > 2$. We have*

$$\int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp\left(-\frac{c_\kappa}{t} d(x, y)^2\right) dt \leq c_{\kappa, s} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s}.$$

Proof. By (2),

$$w(B(x, t^{\frac{1}{2}}))^{-1} \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \begin{cases} \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} & \int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp\left(-\frac{c_\kappa}{t} d(x, y)^2\right) dt \leq \\ & \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N \exp\left(-\frac{c_\kappa}{t} d(x, y)^2\right) dt + \\ & + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_{d(x, y)^2}^\infty t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N dt = \\ & = c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp\left(-\frac{c_\kappa}{t}\right) dt + \\ & + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_1^\infty t^{-\frac{s+N}{2}} dt. \end{aligned}$$

$\square$

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

Lemma 5.3. *Let $N \geq 2$. For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^n$, we have*

$$|(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{d(x, y)^{1-|\alpha|_1-|\beta|_1}}{w(B(x, d(x, y)))}, \quad x, y \in \mathbb{R}^N.$$

Proof. We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^\infty e^{-t\Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y) = \pi^{-\frac{1}{2}} \int_0^\infty h_t(x, y) \frac{dt}{t^{\frac{1}{2}}}.$$

Using Lemma 5.1, we estimate

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{R_{\text{Dunkl},j}})(x, y)| \leq \\ & \leq \frac{c_{\kappa, \alpha, \beta}}{\pi^{\frac{1}{2}}} \int_0^\infty t^{-\frac{|\alpha|_1 + |\beta|_1 + 1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt. \end{aligned}$$

Applying Lemma 5.2 with $s = 1 + |\alpha|_1 + |\beta|_1$, we conclude

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq \\ & \leq \pi^{-\frac{1}{2}} c_{\kappa, \alpha, \beta} \times c_{\kappa, |\alpha|_1 + |\beta|_1 + 1} w(B(x, d(x, y)))^{-1} \times d(x, y)^{1 - |\alpha|_1 - |\beta|_1}. \end{aligned}$$

The assertion follows immediately. $\square$

Lemma 5.4. *For every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

Proof. We have

$$w(Q) \geq c_\theta \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if $x \in Q$, then $|x - c_Q|_\infty \leq \frac{1}{2} \ell(Q)$ and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$w(x) = \prod_{A \subset \{1, \dots, N\}} (\sum_{k \in A} x_k)^{\theta_A} \leq N^{\sum_{A \subset \{1, \dots, N\}} \theta_A} \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

This completes the proof. $\square$

Theorem 5.5. *Let $N \geq 3$. We have*

$$\|M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{L_N(\mathbb{R}^N)}.$$

Proof. It suffices to estimate, for every chamber Γ and for every $g \in G$, the operator $M_{\chi_\Gamma \cdot f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_{\chi_{g\Gamma}}$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, y), \quad x, y \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space $L_2(\mathbb{R}^N, w(x)dx)$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space $L_2(\Gamma, w(x)dx)$.

Let $N_1 = \dim(\text{span}(R))$, $N_2 = N - N_1$ and let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. For a concrete chamber Γ , let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a linear isomorphism given in Theorem 2.2 so that $\Phi(\Gamma) = \Omega$.

It suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w(\Phi^{-1}x)dx)$. Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w_\theta(x)dx)$.

By Lemma 5.3,

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y) \right| \leq c_{\alpha, \beta, \kappa} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(B(x, |x - y|))}, \quad x, y \in \Omega.$$

The other assumptions in Theorem 4.1 follow from Lemma 5.4, Lemma 3.3 and Lemma 3.4. The assertion follows now from Theorem 4.1. $\square$

6. PROOF OF THEOREM 1.1 (i)

Lemma 6.1. *We have*

$$[T_{e_j}, M_f] = M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} U_v,$$

where

$$(\delta_v f)(x) = \frac{f(x) - (U_v f)(x)}{\langle v, x \rangle}, \quad v \in R, \quad x \in \mathbb{R}^N.$$

Proof. Obvious. $\square$

Lemma 6.2. *We have*

$$\|\delta_v f\|_{L_N(\mathbb{R}^N)} \leq c_N \|f\|_{\dot{W}^{1, N}(\mathbb{R}^N)}.$$

Proof. Without loss of generality, $v = e_N$. Set

$$(D_t h)(x) = h(x_1, \dots, x_{N-1}, tx_N).$$

We have

$$\delta_v f = \int_{-1}^1 D_t(\partial_N f) dt.$$

Thus,

$$\begin{aligned} \|\delta_v f\|_{L_N(\mathbb{R}^N)} &\leq \int_{-1}^1 \|D_t(\partial_N f)\|_{L_N(\mathbb{R}^N)} dt = \\ &= \int_{-1}^1 |t|^{-\frac{1}{N}} \|\partial_N f\|_{L_N(\mathbb{R}^N)} dt = \frac{2N}{N-1} \|\partial_N f\|_{L_N(\mathbb{R}^N)}. \end{aligned}$$

$\square$

Lemma 6.3. *For every $N > 1$, we have*

$$\|[B, A]B^{-1}\|_{N, \infty} \leq c_N \|B^{-1}[B^2, A]B^{-1}\|_{N, \infty}.$$

Proof of Theorem 1.1 (i). We prove the assertion under *a priori* restriction $f \in C_c^\infty(\mathbb{R}^N)$. Passing to the case of $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ is standard (see e.g. [refer to our papers](#)).

We have

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] &= [T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} + T_{e_j} [\Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = \\ &= [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} - R_{\text{Dunkl},j} [\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \|[\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.3, we write

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} c_N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Note that

$$\begin{aligned} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} &= \\ &= \sum_{k=1}^N \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] R_{\text{Dunkl},k} + R_{\text{Dunkl},k} [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Combining the estimates in the last two paragraphs, we obtain

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq c'_N \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c'_N \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.1, the fact that Δ_{Dunkl} commutes with U_v and the triangle inequality, we obtain

$$\begin{aligned} \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|M_{\partial_k} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \end{aligned}$$

$$+ \frac{N}{2(N-1)} \sum_{v \in R} |v| \kappa(v) \|M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}.$$

Thus,

$$\begin{aligned} & \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq c_N'' \sum_{k=1}^N \|M_{\partial_k f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ & + c_\kappa \sum_{v \in R} \|M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

The assertion follows now from Theorem 5.5 and Lemma 6.2. $\square$

7. PROOF OF THEOREM 1.1 (ii)

It is likely possible to combine a number of theorems in [9] (with Proposition 9.5 in [9] being the most important) in order to obtain this result. However, we propose a simpler direct proof. The advantage is twofold (a) we are able to use median method directly (b) proofs in [9] are not self-contained (Hytonen refers to GONG-LI-SINKO and, implicitly, to HYTONEN-2026), while our exposition is self-contained. [insert references](#).

7.1. Abstract lower estimate.

Theorem 7.1. *Let $1 \leq j \leq N$. Let $K : \mathbb{R}^N \rightarrow \mathbb{R}$ be such that¹*

$$K(x, y) \geq c_K \frac{y_j - x_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}, \quad x, y \in \mathbb{R}^N, \quad y_j \geq x_j.$$

Let $\mathbf{w} \in A_\infty$ and let V be an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel K . For every real-valued function $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ with $[V, M_f] \in \mathcal{L}_{N,\infty}(\mathbb{R}^N, \mathbf{w}(x)dx)$, we have $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ and

$$\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)} \leq c_{K,\mathbf{w}} \| [V, M_f] \|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, \mathbf{w}(x)dx)}.$$

Lemma 7.2. *Let V be as in the assumptions of Theorem 7.1. Let real-valued $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ be such that the commutator $[V, M_f]$ is bounded on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$. For every $Q \in \mathbb{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that*

$$\begin{aligned} & \|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}, \\ & \frac{1}{\mathbf{w}(Q)} \min_{c \in \mathbb{R}} \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} \leq c_{K,\mathbf{w}} |\langle [V, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|. \end{aligned}$$

Proof. It follows from the assumptions on K that

$$K(x, y) \geq \frac{c_{K,\mathbf{w}}}{\mathbf{w}(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j, \quad Q \in \mathbb{D}.$$

Here, $c_{K,\mathbf{w}}$ is some strictly positive constant.

Let α be the median of f on $3Q$ and β be the median of f on $Q + 3s(Q)e_j$. Assume for definiteness $\alpha \geq \beta$. Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : f(x) \geq \alpha\}, \\ \xi_{Q,k} &= \mathbf{w}(Q)^{-\frac{1}{2}} \chi_{A_k}, \quad \eta_{Q,k} = \mathbf{w}(Q)^{-\frac{1}{2}} \chi_{B_k}, \quad k = 1, 2. \end{aligned}$$

¹Here, c_K is some strictly positive constant.

We have

$$\begin{aligned} & \langle [V, M_f] \xi_{Q,k}, \eta_{Q,k} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} = \\ &= \frac{1}{\mathbf{w}(Q)} \int_{B_k \times A_k} K(x, y) (f(y) - f(x)) \mathbf{w}(x) \mathbf{w}(y) dx dy. \end{aligned}$$

If $x \in B_1$ and $y \in A_1$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If $x \in B_2$ and $y \in A_2$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} & \langle [V, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \geq \\ & \geq \frac{1}{\mathbf{w}(Q)} \int_{B_1 \times A_1} K(x, y) (|f(y) - \alpha| + |\alpha - \beta|) \mathbf{w}(x) \mathbf{w}(y) dx dy \geq \\ & \geq \frac{c_{K,\mathbf{w}}}{\mathbf{w}^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) \mathbf{w}(x) \mathbf{w}(y) dx dy = \\ &= \frac{c_{K,\mathbf{w}} \mathbf{w}(B_1)}{\mathbf{w}^2(Q)} \cdot \left(\int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + |\alpha - \beta| \mathbf{w}(A_1) \right) \geq \\ &= \frac{c_{K,\mathbf{w}}}{2\mathbf{w}(Q)} \cdot \left(\int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + |\alpha - \beta| \cdot \frac{1}{2} \mathbf{w}(Q) \right) \end{aligned}$$

and

$$\begin{aligned} & -\langle [V, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \geq \\ & \geq \frac{1}{\mathbf{w}(Q)} \int_{B_2 \times A_2} K(x, y) |f(y) - \beta| \mathbf{w}(x) \mathbf{w}(y) dx dy \geq \\ & \geq \frac{c_{K,\mathbf{w}}}{\mathbf{w}^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| \mathbf{w}(x) \mathbf{w}(y) dx dy = \\ &= \frac{c_{K,\mathbf{w}} \mathbf{w}(B_2)}{\mathbf{w}^2(Q)} \cdot \int_{A_2} |f(y) - \beta| \mathbf{w}(y) dy \geq \\ & \geq \frac{c_{K,\mathbf{w}}}{2\mathbf{w}(Q)} \int_{A_2} |f(y) - \beta| \mathbf{w}(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} & \|f - \alpha\|_{L_1(3Q, \mathbf{w}(x)dx)} = \int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + \\ & + \int_{A_2} |f(y) - \alpha| \mathbf{w}(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| \mathbf{w}(y) dy \leq \\ & \leq \int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + \int_{A_2} |f(y) - \beta| \mathbf{w}(y) dy \\ & + \int_{A_2} |\alpha - \beta| \mathbf{w}(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| \mathbf{w}(y) dy \leq \\ & \leq \int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + \int_{A_2} |f(y) - \beta| \mathbf{w}(y) dy + |\alpha - \beta| \mathbf{w}(3Q). \end{aligned}$$

Combining the last 3 inequalities, we conclude

$$\frac{c'_{K,\mathbf{w}}}{\mathbf{w}(Q)} \|f - \alpha\|_{L_1(3Q, \mathbf{w}(x)dx)} \leq \max_{k=1,2} |\langle [V, M_f] \xi_{Q,k}, \eta_{Q,k} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|.$$

If

$$|\langle [V, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}| \geq |\langle [V, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta_Q = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\begin{aligned} \frac{c'_{K,\mathbf{w}}}{\mathbf{w}(Q)} \min_{c \in \mathbb{R}} \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} &\leq \frac{c'_{K,\mathbf{w}}}{\mathbf{w}(Q)} \|f - \alpha\|_{L_1(3Q, \mathbf{w}(x)dx)} \leq \\ &\leq |\langle [V, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|. \end{aligned}$$

□

Lemma 7.3. *Let $\mathbf{w} \in A_p(\mathbb{R}^N)$ for $1 < p < \infty$. In the setting of Theorem 7.1, we have,*

$$\begin{aligned} &\left\| \left\{ \frac{1}{m(Q)^p} \min_{c \in \mathbb{R}} \|f - c\|_{L_{\frac{1}{p}}(3Q)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{K,\mathbf{w}} [\mathbf{w}]_{A_p(\mathbb{R}^N)} \| [V, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}. \end{aligned}$$

Proof. Let $\{\xi_Q\}_{Q \in \mathbb{D}}$ and $\{\eta_Q\}_{Q \in \mathbb{D}}$ be functions constructed in Lemma 7.2. By Lemma 2.6,

$$\begin{aligned} &\left\| \left\{ \langle [V, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{\mathbf{w},\infty,7,\infty,7,N,\infty} \| [V, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}. \end{aligned}$$

By Lemma 7.2,

$$\begin{aligned} &\left\| \left\{ \frac{1}{\mathbf{w}(3Q)} \min_{c \in \mathbb{R}} \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{K,\mathbf{w}} \| [V, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}. \end{aligned}$$

Applying Hölder inequality to the functions $|f - c|^{\frac{1}{p}} \mathbf{w}^{\frac{1}{p}}$ and $\mathbf{w}^{-\frac{1}{p}}$, we obtain

$$\int_{3Q} |f(x) - c|^{\frac{1}{p}} dx \leq \left(\int_{3Q} |f(x) - c| \mathbf{w}(x) dx \right)^{\frac{1}{p}} \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{1-\frac{1}{p}}.$$

Taking the latter inequality to the power p , we obtain

$$\|f - c\|_{L_{\frac{1}{p}}(3Q)} \leq \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} \cdot \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{p-1}.$$

Thus,

$$\begin{aligned} &\left\| \left\{ \frac{1}{m(Q)^p} \min_{c \in \mathbb{R}} \|f - c\|_{L_{\frac{1}{p}}(3Q)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{K,\mathbf{w}} \| [V, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} \times \sup_{Q \in \mathbb{D}} \frac{\mathbf{w}(Q)}{m(Q)^p} \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{p-1}. \end{aligned}$$

Replacing the supremum over $\mathbb{D}$ with (an obviously bigger) supremum over all cubes in $\mathbb{R}^N$, we obtain exactly $[\mathbf{w}]_{A_p(\mathbb{R}^N)}$ (see Definition 7.1.3 in [6]). This completes the proof. □

Lemma 7.4. *Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$. For every $p \in (1, \infty)$, we have*

$$\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)} \leq c_{p,N} \left\| \left\{ \frac{1}{m(Q)^p} \min_{c \in \mathbb{R}} \|f - c\|_{L_{\frac{1}{p}}(3Q)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})}.$$

Proof. **supply reference!** □

Proof of Theorem 7.1. Let $p \in (1, \infty)$ be such that $\mathbf{w} \in A_p(\mathbb{R}^N)$. The assertion follows by combining Lemmas 7.3 and 7.4. □

7.2. Proof of Theorem 1.1 (ii).

Lemma 7.5. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\text{Dunkl},j}(x, y) \geq c_\kappa \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{\text{Dunkl},j}(x, y) \leq c_\kappa \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The proof is identical to that in [7, Theorem 1.2]. □

Lemma 7.6. *Let Q be a cube in $\mathbb{R}^N$. We have*

$$\begin{aligned} \sup_{x \in Q} w(x) &\leq N^{\sum_{v \in R} \kappa(v)} \cdot \prod_{v \in R} (|\langle c_Q, v \rangle| + \ell(Q))^{\kappa(v)}, \\ \frac{w(Q)}{m(Q)} &\geq c_\kappa \prod_{v \in R} (|\langle c_Q, v \rangle| + \ell(Q))^{\kappa(v)}. \end{aligned}$$

Proof. If $x \in Q$, then $x = c_Q + y$, $|y| \leq N\ell(Q)$. Thus,

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + N\ell(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + N\ell(Q))^{\kappa(v)}.$$

This immediately yields first the assertion.

We have $B(c_Q, \frac{1}{2}\ell(Q)) \subset Q$. Thus,

$$\frac{w(Q)}{m(Q)} \geq \frac{w(B(c_Q, \frac{1}{2}\ell(Q)))}{m(Q)} = c_N \frac{w(B(c_Q, \frac{1}{2}\ell(Q)))}{m(B(c_Q, \frac{1}{2}\ell(Q)))}.$$

The second assertion follows now from (1). □

Lemma 7.7. *We have $w \in A_\infty(\mathbb{R}^N)$. Thus, $w \in A_p(\mathbb{R}^N)$ for some $1 < p < \infty$.*

Proof. It follows from Lemma 7.6

$$\sup_{x \in Q} w(x) \leq \frac{c_\kappa}{m(Q)} \int_Q w(x) dx$$

holds for every cube in $\mathbb{R}^N$. The assertion follows from Theorem 7.3.3 (c) in [6]. □

Proof of Theorem 1.1 (ii). Since $R_{\text{Dunkl},j}$ is self-adjoint on $L_2(\mathbb{R}^N, w(x)dx)$, it follows that

$$[R_{\text{Dunkl},j}, M_f]^* = -[R_{\text{Dunkl},j}, M_f].$$

Thus,

$$[R_{\text{Dunkl},j}, M_{\Re(f)}], [R_{\text{Dunkl},j}, M_{\Im(f)}] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Hence, we may assume without loss of generality that f is real-valued. It follows from Lemmas 7.5 and 7.7 that the assumptions in Theorem 7.1 are satisfied for $V = R_{\text{Dunkl},j}$ and $\mathbf{w} = w$. The assertion follows now from Theorem 7.1. □

8. PROOF OF SMOOTHNESS

Theorem 8.1. *Let $f \in C_c^\infty(\mathbb{R}^N)$.*

(i) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 = n$, the operator²*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$;

(ii) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 > n$, the operator³*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$;

Lemma 8.2. *Let $f \in C_c^\infty(\mathbb{R}^N)$. We have*

$$[\Delta_{\text{Dunkl}}, M_f] = 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (T_{e_j} U_v + U_v T_{e_j}) + \sum_{g \in G} M_{f_g} U_g.$$

All functions on the right hand side are smooth and compactly supported.

Proof. We have

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f] + [T_{e_j}, M_f] T_{e_j} = \\ &= \sum_{j=1}^N T_{e_j} M_{\partial_j f} + M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (T_{e_j} M_{\delta_v f} U_v + M_{\delta_v f} U_v T_{e_j}) = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) [T_{e_j}, M_{\delta_v f}] U_v = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N M_{\partial_j^2 f} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} \alpha_j \kappa(v) M_{\delta_v(\partial_j f)} U_v + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\partial_j(\delta_v f)} U_v + \frac{1}{4} \sum_{j=1}^N \sum_{v, v' \in R} v_j v'_j \kappa(v) \kappa(v') M_{\delta_{v'} \delta_v f} U_{v'} U_v. \end{aligned}$$

□

Lemma 8.3. *For every $f \in C_c^\infty(\mathbb{R}^N)$, the operator*

$$[\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

is bounded and

$$(1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \in \mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

²Commutators are repeated n times

³Commutators are repeated n times

Proof. We prove only the first assertion. The proof of the second one is similar (modulo Theorem 5.5).

Since each T_{e_j} commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$, each U_v commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$ and since the operators

$$\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, \quad 1 \leq j \leq N,$$

are bounded, the assertion follows from Lemma 8.2. $\square$

Lemma 8.4. *For every $n \in \mathbb{N}$, the operator⁴*

$$[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}$$

is bounded.

Proof. Denote for brevity

$$\text{Expr}_n(f) = [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}.$$

We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, the operator $\text{Expr}_n(f)$ is bounded." Base of induction is given by Lemma 8.3. It suffices to prove the step of induction.

Since each T_{e_j} and each U_v commutes with Δ_{Dunkl} , it follows from Lemma 8.2 that

$$\begin{aligned} \text{Expr}_{n+1}(f) &= 2 \sum_{j=1}^N \text{Expr}_n(\partial_j f) \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \text{Expr}_n(\delta_v f) \cdot \left(\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_v + U_v \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \right) + \\ &+ \sum_{g \in G} \text{Expr}_n(f_g) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_g. \end{aligned}$$

In each summand, the first factor is bounded by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Proof of Theorem 8.1 (i). The assertion follows from Lemma 8.4 and Hadamard 3 lines theorem. $\square$

Lemma 8.5. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-1} \in \mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$\mathcal{D} = \sum_{j=1}^N \gamma_j \otimes T_{e_j},$$

where $\{\gamma_j\}_{j=1}^N$ are Pauli matrices. We have

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= (1 \otimes M_f)(\mathcal{D}^2 + 1)^{-1} = (1 \otimes M_f)(\mathcal{D} + i)^{-1}(\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + [1 \otimes M_f, (\mathcal{D} + i)^{-1}](\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}[\mathcal{D}, 1 \otimes M_f](\mathcal{D}^2 + 1)^{-1} = \end{aligned}$$

⁴Commutators are repeated n times

$$\begin{aligned}
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1} \left(\sum_{j=1}^N \gamma_j \otimes [T_{e_j}, M_f] \right) (\mathcal{D}^2 + 1)^{-1} = \\
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1} (\gamma_j \otimes M_{\partial_j f}) (\mathcal{D}^2 + 1)^{-1} + \\
&\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1} (\gamma_j \otimes M_{\delta_v f} U_v) (\mathcal{D}^2 + 1)^{-1}.
\end{aligned}$$

Since Δ_{Dunkl} commutes with U_v , it follows that

$$\begin{aligned}
&1 \otimes M_f (1 + \Delta_{\text{Dunkl}})^{-1} = \\
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes 1) + \\
&\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes U_v).
\end{aligned}$$

Clearly,

$$\begin{aligned}
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} = \\
&= \|(\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}} M_f (\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&= \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&\leq 2^{\frac{2}{N}} \|M_{|f|^{\frac{1}{2}}} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))}^2 \leq \\
&\stackrel{\text{Th. 5.5}}{\leq} c_{N, \kappa} \| |f|^{\frac{1}{2}} \|_{L_N(\mathbb{R}^N)}^2 = c_{N, \kappa} \|f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}
\end{aligned}$$

and, similarly,

$$\begin{aligned}
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\partial_j f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}, \\
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\delta_v f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}.
\end{aligned}$$

This completes the proof. $\square$

Lemma 8.6. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))."$$

Base of induction (i.e. cases $n = 1$ and $n = 2$) are given by Theorem 5.5 and Lemma 8.5. It suffices to prove the step of induction.

We have

$$\begin{aligned}
&M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = \\
&= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + [M_f, (1 + \Delta_{\text{Dunkl}})^{-1}] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} = \\
&= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}}.
\end{aligned}$$

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

Thus,

$$\begin{aligned} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + \\ &+ (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

In the first summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.5 and the second one belongs to the ideal $\mathcal{L}_{\frac{N}{n-1}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption. In the second summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.3, the second factor belongs to the ideal $\mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the third one is obviously bounded. $\square$

Lemma 8.7. *Let $k, l \in \mathbb{Z}_+$ with $l > k$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. We prove the assertion by induction on k . Base of induction (i.e., the case $k = 0$) is established in Lemma 8.6.

We have

$$\begin{aligned} T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f T_{e_{m_k}}(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} [T_{e_{m_k}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = \\ &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times \frac{T_{e_{m_k}}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\partial_{m_k}} f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ &+ \frac{1}{2} \sum_{v \in R} v_{m_k} \kappa(v) T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\delta_v} f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v. \end{aligned}$$

In each summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Lemma 8.8. *Let $s_1, s_2 \in \mathbb{R}$ be such that $s_1 + s_2 > 0$. We have*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. Step 1: Let us prove the assertion for $s_1, s_2 \in \mathbb{Z}$.

Suppose first $s_1 \leq 0$. Denote $k = -s_1 \geq 0$ and $l = s_2 > k$. Denote for brevity

$$V = (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} (1 + \sum_{j_1, \dots, j_k=1}^N |T_{e_{j_1}} \cdots T_{e_{j_k}}|)^{-1}.$$

Obviously, V is bounded. We have

$$\begin{aligned} (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} &= V \times (1 + \sum_{m_1, \dots, m_k=1}^N |T_{e_{m_1}} \cdots T_{e_{m_k}}|) = V + \\ &+ \sum_{m_1, \dots, m_k=1}^N V \times \text{sgn}(T_{e_{m_1}} \cdots T_{e_{m_k}})^* \times T_{e_{m_1}} \cdots T_{e_{m_k}}. \end{aligned}$$

It, therefore, suffices to prove that operators

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}, \quad T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This is established in Lemma 8.6 and Lemma 8.7, respectively. Hence, the claim is proved for $s_1, s_2 \in \mathbb{Z}$ and $s_1 \leq 0$.

If $s_2 \leq 0$, then the claim follows from that for $s_2 \leq 0$ by taking the adjoint. If $s_1, s_2 \geq 0$, then, according to Lemma 8.7 the operators

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}} M_f, \quad M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. The claim for $s_1, s_2 \geq 0$ follows now from the Hadamard 3 lines theorem.

Step 2: The general case follows now from Step 1 by complex interpolation. $\square$

Proof of Theorem 8.1 (ii). Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

We have

$$\begin{aligned} &(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-s_1}{2}} \times \\ &\quad \times (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1-n}{2}} M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}. \end{aligned}$$

The first factor is bounded by Theorem 8.1 (i). The second factor belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.8. $\square$

9. APPROXIMATION THEOREM

Theorem 9.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} M_{w^{\frac{1}{2}}} [R_{\text{Dunkl}, j}, M_f] M_{w^{-\frac{1}{2}}} &\in V \Delta^{-\frac{1}{2}} + (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N)), \\ V &= M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} U_v - \sum_{k=1}^N R_j M_{\partial_k f} R_k - \\ &\quad - \frac{1}{4} R_j \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} (R_k U_v + U_v R_k). \end{aligned}$$

9.1. First approximation theorem.

Theorem 9.2. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{1}{2}[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in \\ & \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Lemma 9.3. *Let $B \geq 1$. Suppose*

$$B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

We have

$$[B, A]B^{-1} - \frac{1}{2}[B^2, A]B^{-2} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B = \frac{2}{\pi} \int_0^\infty \frac{B^2 d\lambda}{\lambda^2 + B^2}.$$

Hence,

$$[B, A]B^{-1} = \frac{2}{\pi} \int_0^\infty \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} d\lambda.$$

Obviously,

$$\begin{aligned} \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} &= - \left[\frac{\lambda^2}{\lambda^2 + B^2}, A \right] B^{-1} = \frac{\lambda^2}{\lambda^2 + B^2} [B^2, A] \frac{1}{B(\lambda^2 + B^2)} = \\ &= [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} - \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2}. \end{aligned}$$

Obviously,

$$\begin{aligned} \int_0^\infty [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} d\lambda &= [B^2, A] B^{-1} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + B^2)^2} d\lambda = \\ &= [B^2, A] B^{-2} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + 1)^2} d\lambda = \frac{\pi}{4} [B^2, A] B^{-2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty} \left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty. \end{aligned}$$

Next,

$$\left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{\lambda^2 t}{\lambda^2 + t^2} \leq \lambda$$

and

$$\left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{t^{\frac{3}{2}}}{(\lambda^2 + t^2)^2} \leq (\lambda^2 + 1)^{-\frac{5}{4}}.$$

Thus,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \frac{\lambda}{(\lambda^2 + 1)^{\frac{5}{4}}} \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty}. \end{aligned}$$

Hence, the integral

$$\int_0^\infty \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} d\lambda$$

converges absolutely in $\mathcal{L}_{\frac{2N}{3}, \infty}$ and the assertion follows. $\square$

Proof of Theorem 9.2. The assertion follows from Theorem 8.1 and Lemma 9.3. $\square$

9.2. Second approximation theorem.

Theorem 9.4. *If $f \in C_c^\infty(\mathbb{R}^N)$, then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Lemma 9.5. *Let A, B_1, B_2 be such that $B_1, B_2 \geq 1$. Suppose that*

- (i) $A(B_2 - B_1)B_2^{-\frac{1}{2}}, [B_1, A]B_1^{-\frac{1}{2}}$ and $[B_1, A]B_2^{-\frac{1}{2}}$ are bounded;
- (ii) *there exists A' with $B_1^{-\frac{3}{4}}A' \in \mathcal{L}_{\frac{2N}{3}, \infty}$ such that $A = A'A$ and $[B_1, A] = A'[B_1, A]$;*

We have

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B_k^{-\frac{1}{2}} = \frac{2}{\pi} \int_0^\infty \frac{d\lambda}{\lambda^2 + B_k}, \quad k = 1, 2.$$

Thus,

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} = \int_0^\infty A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) d\lambda.$$

Obviously,

$$\begin{aligned} A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) &= A \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_2}. \end{aligned}$$

Consider, for example, the first summand. We write

$$\begin{aligned} \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} &= \frac{1}{\lambda^2 + B_1} A' \cdot A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \cdot B_1^{-\frac{3}{4}} A' \cdot A(B_2 - B_1) B_2^{-\frac{1}{2}} \cdot \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2}. \end{aligned}$$

Thus,

$$\begin{aligned} &\left\| \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} \right\|_{\frac{3N}{4}, \infty} \leq \\ &\leq \left\| \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \right\|_\infty \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \left\| \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2} \right\|_\infty \leq \\ &\leq \sup_{t \geq 1} \frac{t^{\frac{3}{4}}}{\lambda^2 + t} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \sup_{t \geq 1} \frac{t^{\frac{1}{2}}}{\lambda^2 + t} \leq \\ &\leq (\lambda^2 + 1)^{-\frac{1}{4}} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot (\lambda^2 + 1)^{-\frac{1}{2}}. \end{aligned}$$

Hence, the first summand is absolutely integrable in $\mathcal{L}_{\frac{3N}{4}, \infty}$. Similarly, so are the other summands. This completes the proof. $\square$

Lemma 9.6. *Let P be a differential operator of degree 1 with smooth compactly supported coefficients. Suppose that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. The operators*

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}, \quad P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$.

Proof. By assumption, $P = M_{w^{-\frac{1}{2}}} P' M_{w^{\frac{1}{2}}}$, where P' is a differential operator of degree 1 with smooth compactly supported coefficients. Thus,

$$P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = M_{w^{-\frac{1}{2}}} \cdot P'(1 + \Delta)^{-\frac{1}{2}} \cdot M_{w^{\frac{1}{2}}}.$$

Boundedness of the right hand side is obvious.

Similarly,

$$P = \sum_{j=1}^N M_{f_j} T_{e_j} + \sum_{g \in G} M_{h_g} U_g,$$

where functions $\{f_j\}_{j=1}^N$ and $\{h_g\}_{g \in G}$ are smooth and compactly supported. Since the operators $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} = \sum_{j=1}^N M_{f_j} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \sum_{g \in G} M_{h_g} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_g.$$

Boundedness of the right hand side is obvious. $\square$

Proof of Theorem 9.4. Step 1: Suppose first that f is supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$.

We apply Lemma 9.5 with $A = M_f$, $B_1 = 1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}$ and $B_2 = 1 + \Delta_{\text{Dunkl}}$.

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $f = f\phi$ and set $A' = M_\phi$. We have

$$M_\phi[M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f].$$

Thus, $A = A'A$ and $[B_1, A] = A'[B_1, A]$. It follows from the usual Cwikel estimates that $B_1^{-\frac{3}{4}} A' \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This verifies the assumption (ii) in Lemma 9.5.

Note that $[B_1, A]$ is a differential operator P of degree 1 with smooth compactly supported coefficients. Furthermore, the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. By Lemma 9.6, the operators

$$[B_1, A]B_1^{-\frac{1}{2}} = P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \quad [B_1, A]B_2^{-\frac{1}{2}} = P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}},$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$. Note that there exist differential operators $\{P_g\}_{g \in G}$ of degree 1 with smooth compactly supported coefficients such that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ and such that

$$A(B_2 - B_1) = \sum_{g \in G} P_g U_g.$$

Since $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$A(B_2 - B_1) = \sum_{g \in G} P_g (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \cdot U_g$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. This verifies the assumption (i) in Lemma 9.5.

Using Lemma 9.5, we infer the claim of Step 1.

Step 2: Choose a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ such that (a) $0 \leq \phi_n \leq 1$ for $n \geq 1$ (b) ϕ_n is supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, for $n \geq 1$ (c) $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$.

By Step 1,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

for every $n \geq 1$.

By Theorem 5.5,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \rightarrow M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. By the usual Cwikel estimate,

$$M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \rightarrow M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. Thus

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} &= \\ &= M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \end{aligned}$$

where the limit is understood in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since the separable part $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

9.3. Proof of Theorem 9.1.

Lemma 9.7. *We have*

$$[R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$h(x) = 1 - \frac{|x|}{(1 + |x|^2)^{\frac{1}{2}}}, \quad x \in \mathbb{R}^N.$$

We, obviously, have

$$R_{\text{Dunkl},j} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}).$$

Thus,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] &= \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) M_f - M_f h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j} = \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) (1 + \Delta_{\text{Dunkl}}) \cdot (1 + \Delta_{\text{Dunkl}})^{-1} M_f - \\ &\quad - M_f (1 + \Delta_{\text{Dunkl}})^{-1} \cdot (1 + \Delta_{\text{Dunkl}}) h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j}. \end{aligned}$$

In the first summand on the right hand side, the first and the second factors are bounded and the third one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. In the second summand on the right hand side, the second and the third factors are bounded and the first one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. $\square$

Lemma 9.8. *Let $\{B_j\}_{j=1}^N$ and $\{D_j\}_{j=1}^N$ be commuting tuples of self-adjoint operators on $L_2((\mathbb{R})^N, w(x)dx)$. Denote*

$$\langle B \rangle = (1 + \sum_{j=1}^N B_j^2)^{\frac{1}{2}}, \quad \langle D \rangle = (1 + \sum_{j=1}^N D_j^2)^{\frac{1}{2}}.$$

Suppose that there exists a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ with

- (i) $0 \leq \phi_n \leq 1$ for $n \geq 1$ and $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$;
- (ii)

$$M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \quad n \geq 1;$$

- (iii) $M_{\phi_n}(B_k - D_k)$ is bounded⁵ and compactly supported for $n \geq 1$ and $1 \leq k \leq N$;
- (iv) $[D_k, M_{f\phi_n}]$ is bounded⁶ and compactly supported for $n \geq 1$ and $1 \leq k \leq N$;
- (v) $[\langle D \rangle^{-1}, M_{f\phi_n}] \in (\mathcal{L}_{N,\infty})_0(\mathbb{R}^N, w(x)dx)$ for every $n \geq 1$;
- (vi)

$$\|M_h \langle B \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N),$$

$$\|M_h \langle D \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N);$$

We have

$$\frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Choose real-valued $\psi_n \in C_c^\infty(\mathbb{R}^N)$ such that

$$\begin{aligned} M_{\phi_n}(B_k - D_k) &= M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n}, \quad 1 \leq k \leq N, \\ [M_{f\phi_n}, D_k] &= M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n}, \quad 1 \leq k \leq N. \end{aligned}$$

We have

$$\begin{aligned} & \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \frac{B_j}{\langle B \rangle} \cdot M_f (M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1}) \cdot \frac{B_k}{\langle B \rangle} + \\ & + \frac{B_j}{\langle B \rangle} \cdot [M_{f\phi_n}, \langle D \rangle^{-1}] \cdot \frac{B_k}{\langle B \rangle} + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_f \cdot M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n} \langle B \rangle^{-1} + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n} \langle B \rangle^{-1} + \frac{B_j}{\langle B \rangle} \frac{D_k}{\langle D \rangle} \cdot M_f (M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1}) + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [D_k, M_{f\phi_n}] \cdot M_{\psi_n} \langle D \rangle^{-1} + \frac{B_j}{\langle B \rangle} \cdot [\langle D \rangle^{-1}, M_{f\phi_n}] \cdot \frac{D_k}{\langle D \rangle} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot (B_j - D_j) M_{\phi_n} \cdot M_f \frac{D_k}{\langle D \rangle^2} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot [D_j, M_{f\phi_n}] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2} + (\langle B \rangle^{-1} - \langle D \rangle^{-1}) M_{\phi_n} \cdot M_f \frac{D_j D_k}{\langle D \rangle^2} + \\ & + \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_j] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2}. \end{aligned}$$

It follows now from the assumptions and from Hölder inequality that

$$\frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

⁵By a constant which depends also on n .

⁶By a constant which depends also on n .

Obviously,

$$\begin{aligned}
& \left\| \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
& \leq \left\| M_{f(1-\phi_n)} \langle B \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}, \\
& \left\| \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
& \leq \left\| M_{f(1-\phi_n)} \langle D \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}.
\end{aligned}$$

Thus,

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \\
& = \frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2},
\end{aligned}$$

where the limit is taken in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

Lemma 9.9. *For every $f \in C_c^\infty(\mathbb{R}^N)$,*

$$\begin{aligned}
& \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v \in \\
& \in M_{w^{-\frac{1}{2}}} R_j M_f R_k U_v \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \\
& \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f U_v \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\
& \in M_{w^{-\frac{1}{2}}} R_j M_f U_v R_k \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Proof. The assertion follows by applying Lemma 9.8 to the tuples $(T_{e_1}, \dots, T_{e_N})$ and $(M_{w^{-\frac{1}{2}}} \partial_1 M_{w^{\frac{1}{2}}}, \dots, M_{w^{-\frac{1}{2}}} \partial_N M_{w^{\frac{1}{2}}})$ and sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}_{v \in R}$. $\square$

Proof of Theorem 9.1. We have

$$\begin{aligned}
& \left[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f \right] = \\
& = [T_{e_j}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}.
\end{aligned}$$

Using Lemma 9.7 and Theorem 9.2, we write

$$\begin{aligned}
& [R_{\text{Dunkl},j}, M_f] \in [T_{e_j}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \\
& - \frac{1}{2} \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

It follows from Lemma 8.2 and Lemma 8.6 that

$$\begin{aligned}
& [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-1} - 2 \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
& - \frac{1}{2} \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Thus,

$$\begin{aligned}
[R_{\text{Dunkl},j}, M_f] &\in M_{\partial_j f}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v - \\
&\quad - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
&\quad - \frac{1}{4} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in \\
&\quad \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

The assertion follows from Lemma 9.9. $\square$

10. SPECTRAL ASYMPTOTIC FORMULA

Theorem 10.1. *Let $N > 2$. Let Π be the C^* -algebra of operators admitting principal symbol. For every $T \in \Pi \rtimes_{\alpha} G$ compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, T \Delta^{-\frac{1}{2}}) = c_N \|(\text{sym} \rtimes G)(T)\|_{L_N((L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G)}.$$

Definition 10.2. *Let G be a Weyl group, acting orthogonally on $\mathbb{R}^N$, and let Γ be a fixed chamber. Let $\{e_g\}_{g \in G}$ be the standard basis of $\mathbb{C}^G$. Define a surjective partial isometry $W : L_2(\mathbb{R}^N) \otimes \mathbb{C}^G \rightarrow L_2(\mathbb{R}^N)$ by setting*

$$W \left(\sum_{g \in G} \xi_g \otimes e_g \right) = \sum_{g \in G} (\xi_g|_{\Gamma}) \circ g^{-1}.$$

The crucial observation is

$$\mu(W^* T W) = \mu(T), \quad T \in B(L_2(\mathbb{R}^N)).$$

Indeed, this is true for every surjective isometry.

Lemma 10.3. *If $f \in C_c^{\infty}(\mathbb{R}^N)$, then*

$$M_f \Delta^{-\frac{1}{2}} \in \sum_{g \in G} M_{f \chi_{g\Gamma}} \Delta^{-\frac{1}{2}} M_{\chi_{g\Gamma}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)).$$

Proof. This follows from Theorem 4.1 in [5]. $\square$

Lemma 10.4. *If $f \in C_c^{\infty}(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^{\infty}(\mathbb{S}^{N-1})$, then*

$$h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \in \sum_{g \in G} M_{\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} + \mathcal{K}(L_2(\mathbb{R}^N)).$$

Proof. Indeed, $f \chi_{g\Gamma} \in C_c^{\infty}(\mathbb{R}^N)$ for $g \in \Gamma$. Hence, **this requires a reference!!**

$$[h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f \chi_{g\Gamma}}] \in \mathcal{K}(L_2(\mathbb{R}^N)), \quad g \in G.$$

The assertion follows now from the following algebraic identity.

$$h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f = \sum_{g \in G} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} = \sum_{g \in G} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} \cdot M_{\chi_{g\Gamma}} =$$

$$\begin{aligned}
&= \sum_{g \in G} M_{f\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}} = \\
&= \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot M_{f\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}} = \\
&= \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f\chi_{g\Gamma}} \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot [M_{f\chi_{g\Gamma}}, h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)] \cdot M_{\chi_{g\Gamma}} + \\
&\quad + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}}.
\end{aligned}$$

□

Lemma 10.5. *Let $f \in C_c^\infty(\mathbb{R}^N)$ and let $g \in G$. We have*

$$\begin{aligned}
W^* M_f W &= \sum_{g \in G} M_{(f\chi_{g\Gamma}) \circ g} \otimes E_{g,g}, \\
W^* (M_{\chi_{g\Gamma}} \Delta^{-\frac{1}{2}} M_{\chi_{g\Gamma}}) W &= M_{\chi_\Gamma} \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} \otimes E_{g,g}, \\
W^* M_{\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{\chi_{g\Gamma}} W &= M_{\chi_\Gamma} (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{\chi_\Gamma} \otimes E_{g,g}, \\
W^* U_g W &= \sum_{g' \in G} M_{\chi_\Gamma} \otimes E_{gg',g'}.
\end{aligned}$$

Proof. The third assertion follows from the equality

$$U_g^{-1} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) U_g = (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right).$$

The other assertions are obvious. □

Lemma 10.6. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$\begin{aligned}
&W^* h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \Delta^{-\frac{1}{2}} W \in \\
&\in \sum_{g \in G} \left[M_{\chi_\Gamma} (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{(f\chi_{g\Gamma}) \circ g} \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} \right] \otimes E_{g,g} + (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N) \otimes \mathbb{C}^G).
\end{aligned}$$

Proof. We write $f = f_1 f_2$, where $f_1, f_2 \in C_c^\infty(\mathbb{R}^N)$ are supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. The assertion follows from Lemma 10.3 (applied with f_2 in place of f), Lemma 10.4 (applied with f_1 in place of f) and Lemma 10.5. □

Lemma 10.7. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$\begin{aligned}
&W^* h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \Delta^{-\frac{1}{2}} U_g W \in \\
&\in \sum_{g' \in G} \left[(h \circ g') \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-\frac{1}{2}} \right] \otimes E_{g',g^{-1}g'} + (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N) \otimes \mathbb{C}^G).
\end{aligned}$$

Proof. The assertion follows from Lemma 10.6 and the facts (see the proof of Lemma 10.4 for the first one and Theorem 4.1 in [5] for the second one) that

$$M_{\chi_\Gamma}(h \circ g)\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)M_F - (h \circ g)\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)M_F \in \mathcal{K}(L_2(\mathbb{R}^N)),$$

$$M_F\Delta^{-\frac{1}{2}}M_{\chi_\Gamma} - M_{F\chi_\Gamma}\Delta^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)),$$

for every $F \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. $\square$

Theorem 10.8. *Let $N > 2$. Let Π be the C^* -algebra of operators admitting principal symbol. For every $S \in \Pi \otimes M_m(\mathbb{C})$, compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, S(\Delta^{-\frac{1}{2}} \otimes 1)) = c_N \|(\text{sym} \otimes \text{id})(S)\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_m(\mathbb{C}))}.$$

Lemma 10.9. *Let $A = L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})$. Recall that G acts on A by the formula*

$$(\beta(g))(f \otimes 1) = (f \circ g^{-1}) \otimes 1, \quad (\beta(g))(1 \otimes h) = 1 \otimes (h \circ g^{-1}).$$

There is $$ -homomorphism $\Phi : A \rtimes_\beta G \rightarrow A \otimes M_G(\mathbb{C})$ such that*

$$\Phi(aU_g) = \sum_{g' \in G} (\beta(g'^{-1}))(a) \otimes E_{g', g^{-1}g'}, \quad a \in A, \quad g \in G.$$

We have

$$\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \otimes \text{Tr} \right) \circ \Phi = |G| \cdot \left(\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \right) \rtimes_\beta G \right).$$

the proof is available for convenience of co-authors. however, it is too elementary to be included into the text. so, it is commented out.

Lemma 10.10. *Let $X \in A \rtimes_\beta G$. Let*

$$V_g = \sum_{g' \in G} E_{g'g^{-1}, g} \in M_G(\mathbb{C})$$

be a unitary matrix. We have

$$(\beta(g) \otimes \text{id})(\Phi(X)) = (1 \otimes 1 \otimes V_g^{-1})\Phi(X)(1 \otimes 1 \otimes V_g).$$

the proof is available for convenience of co-authors. however, it is too elementary to be included into the text. so, it is commented out.

Proof of Theorem 10.1. Step 1: Suppose first

$$T = \sum_{g \in G} \sum_{k=1}^K h_{k,g} \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{f_{k,g}} U_g$$

with $h_{k,g} \in C^\infty(\mathbb{S}^{N-1})$ and with $f_{k,g} \in C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$.

Using Lemma 10.7, we write

$$W^* T \Delta^{-\frac{1}{2}} W \in S(\Delta^{-\frac{1}{2}} \otimes 1) + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N) \otimes \mathbb{C}^G),$$

where

$$S = \sum_{g, g' \in G} \sum_{k=1}^K \left[(h_{k,g} \circ g') \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{(f_{k,g} \chi_{g'\Gamma}) \circ g'} \right] \otimes E_{g', g^{-1}g'}.$$

Obviously, $S \in \Pi \otimes M_G(\mathbb{C})$ is compactly supported from the right. We have

$$\begin{aligned} (\text{sym} \otimes \text{id})(S) &= \sum_{g, g' \in G} \sum_{k=1}^K \left[((f_{k,g} \chi_{g'\Gamma}) \circ g') \otimes (h_{k,g} \circ g') \right] \otimes E_{g', g^{-1}g'} = \\ &\stackrel{L.10.9}{=} \Phi \left(\sum_{g \in G} \sum_{k=1}^K (f_{k,g} \otimes h_{k,g}) \right) \cdot (\chi_\Gamma \otimes 1_{C(\mathbb{S}^{N-1})} \otimes 1_{M_G(\mathbb{C})}). \end{aligned}$$

Using Theorem 10.8 and Lemma 2.7, we conclude that

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, T \Delta^{-\frac{1}{2}}) &= \lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, W^* T \Delta^{-\frac{1}{2}} W) = \lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, S(\Delta^{-\frac{1}{2}} \otimes 1)) = \\ &= c_N \left\| \Phi \left(\sum_{g \in G} \sum_{k=1}^K (f_{k,g} \otimes h_{k,g}) \right) \cdot (\chi_\Gamma \otimes 1_{C(\mathbb{S}^{N-1})} \otimes 1_{M_G(\mathbb{C})}) \right\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_G(\mathbb{C}))}. \end{aligned}$$

For every $X \in L_N((L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G)$, we have

$$\begin{aligned} &\left\| \Phi(X) \cdot (\chi_\Gamma \otimes 1_{C(\mathbb{S}^{N-1})} \otimes 1_{M_G(\mathbb{C})}) \right\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_G(\mathbb{C}))} = \\ &\stackrel{L.10.10}{=} |G|^{-\frac{1}{N}} \left\| \Phi(X) \right\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_G(\mathbb{C}))} = \\ &\stackrel{L.10.9}{=} \|X\|_{L_N((L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G)}. \end{aligned}$$

This completes the proof for our special T .

Step 2: Consider now the general T . There exists (see e.g. the proof of Theorem 3.5 in [?]) a sequence $T_K \rightarrow T$ in $\Pi \rtimes_\alpha G$ such that

$$T_K = T'_K + \sum_{g \in G} \sum_{k=1}^K h_{K,k,g} \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{f_{K,k,g}} U_g, \quad 1 \leq k \leq K.$$

Here, $h_{K,k,g} \in C^\infty(\mathbb{S}^{N-1})$, $f_{K,k,g} \in C_c^\infty(\mathbb{R}^N)$ are supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$, $T'_K \in \mathcal{K}(L_2(\mathbb{R}^N))$ is compactly supported from the right. The assertion follows now from Step 1 and Lemmas 2.7, 2.8. $\square$

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